Spacing Out: On the Reliability of Binaural Music Source Separation Metrics

Richa Namballa [1] Magdalena Fuentes[1, 2]

[1]Music and Audio Research Laboratory, New York University, New York, USA

[2]Integrated Design & Media, New York University, New York, USA

Background

- Recent work [1] shows stereo music source separation (MSS) models degrade spatial cues when applied to binaural audio, but this conclusion relies on objective spatial metrics.

Are existing spatial metrics perceptually reliable and robust to MSS artifacts when evaluating binaural music?

Goals

- Retrain an MSS architecture on binaural music data (Bi-SCNet) to compare it against its stereo counterpart (SCNet).
- Conduct a perceptual listening study assessing overall stem quality and spatial localization accuracy.
- Evaluate the correlation between human perceptual judgments and objective spatial distortion metrics.
- Stress-test metric stability against separation noise by source.

Methodology

Model Training

- Architecture: Sparse Compression Network (SCNet) [2]
- Dataset: Binaural-MUSDB [1]
- **Bi-SCNet** trained without channel flipping/scaling augmentations to preserve spatial cues.

Objective Metrics

- ΔITD = Change in Interaural Time Difference (ITD) [3]
- ΔILD = Change in Interaural Level Difference (ILD) [3]
- SSR = Signal to Spatial Distortion Ratio [4]
- SRR = Signal to Residual Distortion Ratio [4]

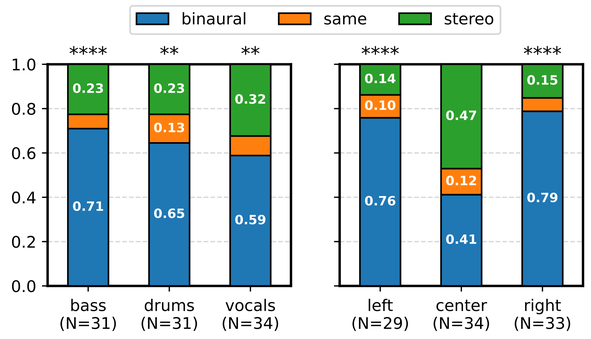
GCC-PHAT for ITD Estimation: Weighted [5] & Masked [6]

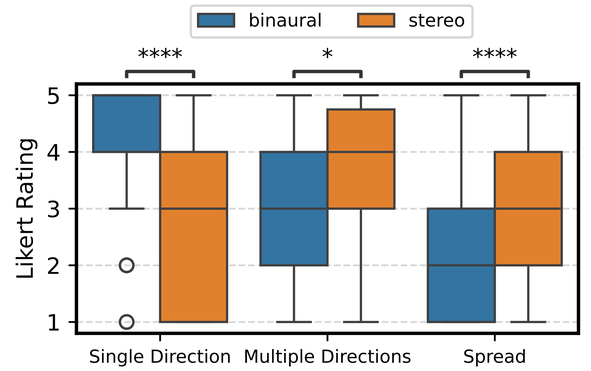
$$\mathbf{c}(\mathbf{s}) = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\mathbf{s}_L) \odot \mathcal{F}(\mathbf{s}_R)^*}{|\mathcal{F}(\mathbf{s}_L) \odot \mathcal{F}(\mathbf{s}_R)^*|^{\beta}}\right)$$

$$M(m) = \begin{cases} 0 & \text{if } |R(m)| < \delta_T \\ 1 & \text{otherwise} \end{cases} \qquad \mathbf{c}(\mathbf{s}) = \mathcal{F}^{-1}\left(\frac{\mathbf{M} \odot \mathcal{F}(\mathbf{s}_L) \odot \mathcal{F}(\mathbf{s}_R)^*}{|\mathcal{F}(\mathbf{s}_L) \odot \mathcal{F}(\mathbf{s}_R)^*|}\right)$$

Perceptual Listening Study

- 26 human participants for online listening study (headphones required).
- Evaluated separation quality of Bi-SCNet and SCNet on binaural mixtures via pairwise comparisons [7] against reference stems.
- Evaluated localization ability with 5-point Likert scale.





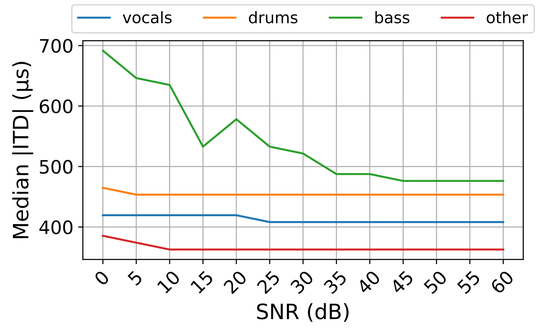
Metric Reliability & Agreement

- SRR and ΔILD align most consistently with human preferences.
- ΔITD shows weakest agreement with perception of source localization.
- SSR shows weaker agreement in the bass compared to other sources.

Source Position & Instrument Sensitivity

- **Centered Sources**: Humans struggle to distinguish stereo vs. binaural model outputs for centered sources.
- **Narrow-Band Instruments**: ΔITD performs especially poorly on low-frequency and narrow-band stems, like the bass.

Fragility of ITD Estimation



- ITD calculation relies on cross-correlation (GCC-PHAT), which collapses in the presence of minor residual interference and separation artifacts when separating the bass (narrow-band).
- Robustness vs. accuracy trade-offs remain even when exploring alternative ITD estimation methods.

Conclusion

- Objective spatial metrics borrowed from speech and other audio do not cleanly translate to binaural music source separation.
- Current metrics like SRR and ΔILD reflect overall loudness balance and noise rather than true spatial position preservation.
- **Key Takeaway**: Future binaural MSS research requires perceptually grounded, artifact-robust spatial metrics designed specifically for complex musical signals.

References

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